DAY - 4

RADIUS & TACACS+ : TACACS+ & RADIUS Features, RADIUS Authentication Process, TACACS+ Use Cases, RADIUS vs. TACACS+

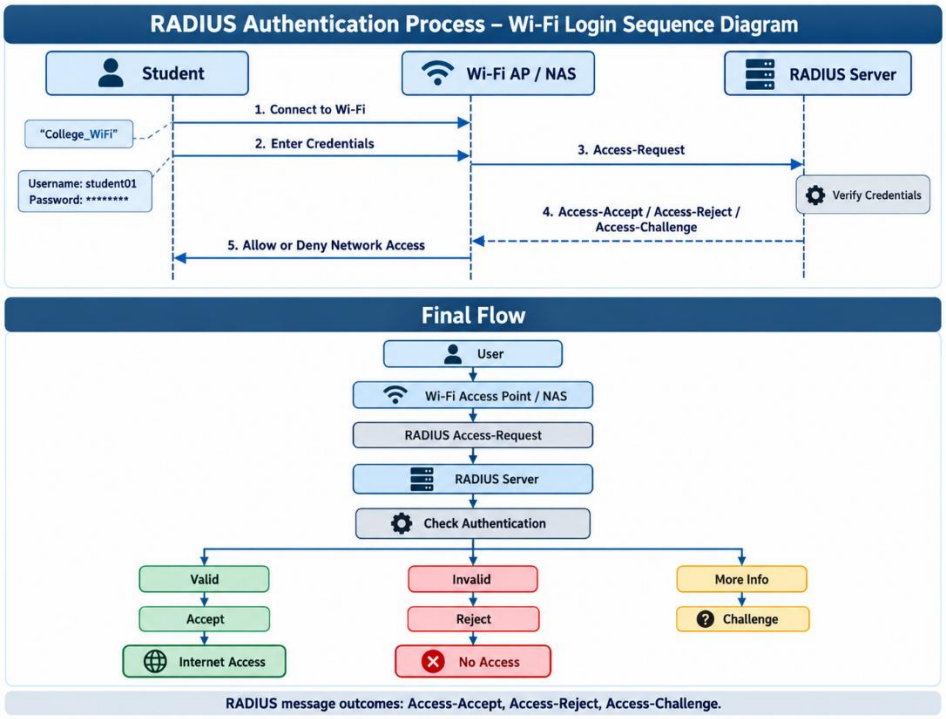
1. Create a comparison table listing at least 4 key differences between RADIUS and TACACS+ features, focusing on protocol type, encryption, ports used, and command authorization.

ANS:

Feature	**RADIUS**	**TACACS+**
Full Form	Remote Authentication Dial-In User Service	Terminal Access Controller Access-Control System Plus
Transport Protocol	UDP	TCP
Port Used	UDP 1812 for Authentication/Authorization and UDP 1813 for Accounting	TCP 49
Encryption / Protection	Traditionally protects the user password, but not the complete RADIUS packet	Protects the packet body, while the TACACS+ header remains visible
AAA Operation	Authentication and authorization are closely combined	Authentication, authorization and accounting can be handled separately
Command Authorization	Not primarily designed for detailed command-by-command authorization	Supports granular command authorization
Typical Usage	Wi-Fi, VPN, hotspot, 802.1X and general network-access authentication	Administrative access to routers, switches, firewalls and other network devices
Best Suited For	Large numbers of network users	Network administrators

2. Draw a simple flowchart or sequence diagram that illustrates the RADIUS authentication process for a user trying to log in to a Wi-Fi network (like at a coffee shop or college campus).

ANS :



RADIUS Authentication Steps

Step 1 – User connects to Wi-Fi

The user selects the college or public Wi-Fi network.

Example:

College_WiFi

Step 2 – User provides authentication information

The user enters credentials or otherwise participates in the authentication method used by the network.

Example:

Username: student01

Password: ********

Step 3 – Network device sends Access-Request

The Wi-Fi access point, wireless controller or Network Access Server (NAS) sends a **RADIUS Access-Request** to the RADIUS server.

Wi-Fi NAS → RADIUS Server

Access-Request

Step 4 – RADIUS server verifies the user

The RADIUS server checks the supplied authentication information against its configured identity database or authentication system.

The RADIUS server can respond with messages such as:

Access-Accept

Access-Reject

Access-Challenge

These message types are part of the RADIUS authentication process.

Step 5 – Network access is granted or denied

If authentication succeeds:

Access-Accept

↓

User gets network access

If authentication fails:

Access-Reject

↓

User is denied network access

3. Describe a real-world scenario from any app you use (such as Zomato, Paytm, or IRCTC) where TACACS+ would be more suitable than RADIUS for managing admin access, and explain why.

ANS:

Example: Paytm Administration System

Consider a **hypothetical Paytm data-center network**. Paytm customers would not use TACACS+ to log into the Paytm mobile application.

Instead, TACACS+ could be used internally for **network administrators who manage routers, switches and firewalls**.

For example, suppose there are three administrators:

Network Administrator

Security Administrator

Junior Network Engineer

They should not necessarily have permission to execute the same commands.

Example

A senior network administrator may be allowed to execute:

configure terminal

interface GigabitEthernet0/1

shutdown

router ospf 1

A junior engineer may only be allowed to execute monitoring commands such as:

show ip interface brief

show running-config

show ip route

ping

With TACACS+, authorization policies can be used to control which administrative commands a particular user is permitted to execute. Command authorization is an established TACACS+ use case.

Why TACACS+ is Suitable

TACACS+ is suitable for this situation because it provides:

1. Centralized authentication

Administrators can authenticate using a centralized AAA server.

2. Granular command authorization

Different administrators can be permitted to use different commands.

Example:

Junior Admin → show commands only

Senior Admin → configuration commands

3. Accounting

Administrative actions can be recorded, making it easier to determine which administrator executed particular actions.

4. Separation of AAA functions

TACACS+ separates authentication, authorization and accounting functions, providing flexible administrator access control.

Therefore, TACACS+ would be a suitable choice for controlling **administrative access to the company's network infrastructure**.

4. List two advantages and two disadvantages of using RADIUS for network authentication in a public hotspot (like those found in malls or airports).

ANS :

Consider a public Wi-Fi network at:

Airport

Mall

Coffee Shop

College

Hotel

Advantages

Advantage 1 – Centralized Authentication

RADIUS allows the organization to maintain authentication centrally.

Instead of every access point maintaining a separate user database:

AP1 ─┐

AP2 ─┼────→ RADIUS Server

AP3 ─┤

AP4 ─┘

All access points or network access devices can rely on the centralized RADIUS infrastructure.

This makes user management easier.

Advantage 2 – Supports Accounting and Large Numbers of Users

RADIUS can support authentication and accounting for many network users.

The system can maintain information associated with sessions, such as:

User login

Session start

Session stop

Connection information

Accounting information

RADIUS accounting is specifically designed to carry accounting information between a NAS and an accounting server.

This makes it useful for environments such as:

Airports

Universities

Hotels

Enterprise Wi-Fi

Public hotspots

Disadvantages

Disadvantage 1 – Limited Native Packet Protection

Traditional RADIUS does not encrypt the complete RADIUS packet. For example, the RFC-defined User-Password attribute is protected using the shared secret and request authenticator, rather than the entire packet being encrypted.

Therefore, additional security mechanisms and secure network design are important when deploying RADIUS.

Disadvantage 2 – Dependency on the Authentication Infrastructure

If the RADIUS server becomes unavailable and no backup or fallback mechanism exists, new users may be unable to authenticate.

Example:

User

↓

Wi-Fi AP

↓

RADIUS Server

X

Server unavailable

↓

Authentication fails

For this reason, organizations commonly design authentication systems with redundancy

or backup servers.

Summary

Advantages	Disadvantages
Centralized authentication	Limited native protection of the overall traditional RADIUS packet
Supports scalable authentication and accounting	RADIUS server/network failure can affect authentication

5. Use ChatGPT or another AI tool to generate a sample TACACS+ configuration for a Cisco router, then review the output and highlight any commands you would modify for a college lab setup. Hint : Ask the AI tool for 'basic TACACS+ configuration for Cisco router' and check for lines like 'tacacs-server host' and 'aaa new-model'.

ANS :

The following is a basic example of TACACS+ configuration for a Cisco router.

Example Network

```
Admin PC
192.168.10.20
    |
    |
+--------------+
| Cisco Router |
|192.168.10.1 |
+--------------+
    |
    |
+----------------+
| TACACS+ Server |
|192.168.10.10 |
+----------------+
```

Basic Configuration

Enter privileged EXEC mode:

Router> enable

Enter configuration mode:

Router# configure terminal

Step 1 – Enable AAA

Router(config)# aaa new-model

The aaa new-model command enables Cisco AAA functionality. Cisco's TACACS+ configuration documentation uses this as the starting point for AAA configuration.

Step 2 – Create a Local Backup Administrator

Router(config)# username labadmin privilege 15 secret Lab@123

This account provides local authentication if the TACACS+ server becomes unavailable.

Step 3 – Configure the TACACS+ Server

For a college lab example:

Router(config)# tacacs-server host 192.168.10.10

Router(config)# tacacs-server key College@123

Here:

192.168.10.10 = TACACS+ Server IP

College@123 = Shared TACACS+ Key

Cisco documentation shows the same basic legacy configuration pattern using tacacs-server host and tacacs-server key.

Step 4 – Configure TACACS+ Login Authentication

Router(config)# aaa authentication login default group tacacs+ local

Meaning:

First → Try TACACS+ Server

↓

If unavailable

↓

Use Local Database

The local fallback is important in a lab because it reduces the risk of being locked out of the router if the TACACS+ server cannot be reached. Cisco documents this TACACS+-plus-local authentication arrangement.

Step 5 – Configure EXEC Authorization

Router(config)# aaa authorization exec default group tacacs+ local

This allows authorization of administrative EXEC sessions.

Step 6 – Configure Command Authorization

For privilege-level 15 commands:

Router(config)# aaa authorization commands 15 default group tacacs+ local

This allows the TACACS+ server to make authorization decisions about privileged commands.

TACACS+ supports authorization requests containing shell commands, which is one of its major advantages for administrator access.

Step 7 – Enable Accounting

Router(config)# aaa accounting exec default start-stop group tacacs+

Router(config)# aaa accounting commands 15 default start-stop group tacacs+

This can record administrator sessions and privileged command activity through TACACS+ accounting.

Step 8 – Configure VTY Access

Router(config)# line vty 0 4

Router(config-line)# login authentication default

Router(config-line)# transport input ssh

Router(config-line)# exit

SSH should be preferred instead of Telnet for remote administrative connections.

Complete TACACS+ Configuration

enabl

configure terminal

aaa new-model

username labadmin privilege 15 secret Lab@12

tacacs-server host 192.168.10.10

```
tacacs-server key College@12
aaa authentication login default group tacacs+ local
aaa authorization exec default group tacacs+ local
aaa authorization commands 15 default group tacacs+ local
aaa accounting exec default start-stop group tacacs+
aaa accounting commands 15 default start-stop group tacacs
line vty 0 4
 login authentication default
 transport input ssh
exit
end
```

Commands I Would Modify for a College Lab

If an AI tool generated a generic TACACS+ configuration, I would modify the following commands according to the college lab topology.

1. TACACS+ Server IP Address

Generic configuration:

```
tacacs-server host 10.2.3.4
```

College lab configuration:

```
tacacs-server host 192.168.10.10
```

Reason: The command must contain the actual IP address assigned to the college lab TACACS+ server.

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2. TACACS+ Shared Key

Generic:

```
tacacs-server key cisco
```

College lab:

```
tacacs-server key College@123
```

Reason: The shared key configured on the Cisco router must match the key configured for that device on the TACACS+ server.

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3. Local Backup Username

Generic:

username admin privilege 15 secret admin

College lab:

username labadmin privilege 15 secret Lab@123

Reason: A local backup administrator should be available in case the TACACS+ server cannot be reached.

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4. Authentication Method

Recommended college lab configuration:

aaa authentication login default group tacacs+ local

I would keep local as a fallback because students may incorrectly configure the server, IP addressing or connectivity during practical exercises.

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5. Remote Access

Instead of:

transport input telnet

I would use:

transport input ssh

because SSH provides encrypted remote management.

Final College Lab Configuration

```
Router> enable
Router# configure termina
Router(config)# aaa new-mode
Router(config)# username labadmin privilege 15 secret Lab@12
Router(config)# tacacs-server host 192.168.10.10
Router(config)# tacacs-server key College@123
Router(config)# aaa authentication login default group tacacs+ local
Router(config)# aaa authorization exec default group tacacs+ local
Router(config)# aaa authorization commands 15 default group tacacs+ loca
Router(config)# aaa accounting exec default start-stop group tacacs+
Router(config)# aaa accounting commands 15 default start-stop group tacacs
Router(config)# line vty 0 4
Router(config-line)# login authentication default
Router(config-line)# transport input ssh
Router(config-line)# exit
Router(config)# end
Router# write memory
```